# Ryan Phua

***Based in Los Angeles/Orange, CA ▪ (949) 478 4109 ▪ Email ▪ LinkedIn ▪ Portfolio ▪ Cover Letter ▪ Open to Relocation ✓***

## Full Stack Application Engineer

**Google Cloud & Amazon Web Services | Automation Scripting | DevOps & Containerization**

### [RELATED COURSES]

- Distributed Systems
- Computer Networking
- Machine-Learning (NLP)
- Operating Systems (OS)

### [AT-A-GLANCE]

- Proficient in cloud-based CI/CD automation, developing scalable, cost-efficient cloud infrastructures & pipelines
- Skilled in Docker and Kubernetes for containerizing & orchestrating microservices

### [TECHNICAL SKILLS]

***<CLOUD & DEVOPS/>***

- AWS: EC2, Lambda, DynamoDB, S3, Amplify, CloudFormation, Terraform, CodeGen, CodePipeline
- Google Cloud
- Docker/Kubernetes

***<AUTOMATION & CI/CD/>***

- CI/CD Pipelining
- Automation Scripting (Linux, zsh, vim, nano)
- Prometheus/Grafana
- Elasticsearch (SEO)

***<FULL STACK APP DEV/>***

- Frontend (React/Next.js, Flutter, Angular)
- Backend (Node, Express, Django, Spring, APIs, REST, GraphQL)
- Databases (PostgreSQL, MySQL, MongoDB)
- Versioning (Git, Geritt)

### [AFFILIATIONS]

- The Daily Bruin | UCLA
- Campus Events Commission | UCLA

### [CERTIFICATIONS]

- AWS Certified Cloud Practitioner (2024)
- AWS Solutions Architect Associate (2025 exp.)
- HarvardX Professional Certificate in Data Science (2021)

### [LANGUAGES]

- English (Advanced)

### [EDUCATION]

- **University of California, Los Angeles (UCLA), Henry Samueli School of Engr & Applied Sci (HSSEAS)**
  *~ Ranked 11th globally for computer science, 7th in the US & 1st among US public colleges*
- **Bachelor of Science (B.S) in Computer Science.** *(exp degree completion by Dec 2025)*

### [WORK EXPERIENCE]

**Full Stack App Development | Software Engineer** **[Jun '24 – Sep '24]**

**Solarbrio Inc. (CA) |** *a renewable energy startup leveraging geospatial mapping tools for resource analysis*

- Managed the successful end-to-end migration of a core product feature from AWS Amplify V1 to V2, building and maintaining full-stack systems to ensure reliability and performance
- Resolved critical versioning issues by replacing deprecated AWS dependencies with custom GraphQL endpoints via AWS AppSync
- Enhanced API performance by integrating a caching layer with Amazon ElastiCache (Redis) at the resolver level to reduce DynamoDB load and latency

  ***Key Achievements*:**
  - Migration ensured seamless CI/CD integration, future Infrastructure-as-Code (IaC) compatibility, 20% faster deployments and 45% reduction in AWS resource costs

**Backend Web Services | Developer** **[Sep '23 – Jul '25]**

**Campus Events Commission (CEC) | University of California, Los Angeles (UCLA)**

- Streamlined development environment via Docker & Kubernetes orchestration, reduced new developer onboarding overhead by 75% (cut allocated onboarding time from 3 to 1 week)
- Integrated Redis caching for API endpoints, cut query load on main database by 30%, boosted response latency by 20% for high throughput API endpoints

  ***Key Achievements*:**
  - Fast-tracked deliverable timeline by 3 weeks overall, initiated and facilitated Scrum sprints on Jira as Backend Lead

**DevOps and Pipeline Engineering | Developer** **[~250 hours]**

**The Daily Bruin |** *UCLA's official student newspaper*

- Completed migration of backend from legacy DigitalOcean droplets to Firebase, dockerized multiple microservices, maintained 100% feature parity post-migration with comprehensive unit/integration testing suite in Jest
- Introduced contract testing using Pact within the CI/CD pipeline to ensure compatibility between microservices and the frontend application

  ***Key Achievements*:**
  - Upgraded CI/CD pipeline from a locally hosted Github repo to a cloud deployment via GitHub Actions & AWS, reducing build times by half and cutting deployment failures by 25%

### [PROJECTS]

**Full Stack Web Development: BruinCreates Artwork Exchange and Marketplace** **[~100 hrs]**

- Directed the frontend engineering team in UI development using React.js and Redux for state management; collaborated with backend leaders to integrate Java Spring REST API endpoints
- Implemented a core seller feature by integrating Instagram's Basic Display and Graph APIs, enabling sellers to dynamically render their art feeds directly on the platform

  ***Key Achievements*:**
  - Architected and delivered a high-performance search engine using Java Spring and Elasticsearch, optimizing indexing strategies to cut search latency by 50% on ES Rally

**Full Stack in Typescript: Social Media Music App with Live Chat** **[~168 hrs]**

- Collaborated with product leaders to translate user requirements and Figma UX wireframes into a viable MVP, focusing on core features for music discovery and social interaction
- Optimized MongoDB schema and simplified backend queries in Typescript to improve execution time by 60%, ensuring consistent throughput during peak load times

  ***Key Achievements*:**
  - 3x decrease in API query latency post database refactoring on AWS CloudWatch metrics